

Jie-En (Matthew) Yao

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Education

University of Southern California

PhD in Computer Science; GPA: 3.95/4.00

Aug 2023 – May 2028

Los Angeles, CA

- Core Areas: Machine Learning, Deep Learning, Computer Vision, Image Processing, Generative Models, Medical AI

National Tsing Hua University

Bachelor of Science in Computer Science; GPA: 4.27/4.30; Class Rank: 1/142

Sep 2018 – Jun 2022

Hsinchu, Taiwan

Research Experience

University of Southern California and China Medical University Hospital

Research Assistant

Jan 2025 – Jan 2026

Los Angeles, CA

- **Topic: Coronary Angiography Interpretation and Stenosis Estimation**

- * Developed a clinically interpretable automated pipeline for medical image analysis
- * Implemented stenosis estimation algorithms to provide accurate vessel narrowing assessments, supporting cardiologists in data-driven clinical decision-making

MediaTek, Intelligence Vision Processing Team

AI Research Intern

Jan – Dec 2022

Hsinchu, Taiwan

- **Topic: Super-Resolution**

- * Prototyped, optimized, and iterated on advanced deep learning architectures, efficiently translating research ideas into high-quality experimental results
- * Developed a novel arbitrary-scale super-resolution model, achieving state-of-the-art performance and publication at CVPR

Publications

Jie-En Yao, et al., "HCL-FF: Hierarchical and Contrastive Learning for Forward-Forward Algorithm", accepted by IEEE Conf. on Computer Vision and Pattern Recognition (CVPR), 2026

Jintang Xue, **Jie-En Yao, et al., "Descrip3D: Enhancing Large Language Model-based 3D Scene Understanding with Object-Level Text Descriptions"**, accepted by IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026

Jie-En Yao, et al., "Local Implicit Normalizing Flow for Arbitrary Scale Image Super-Resolution", accepted by IEEE Conf. on Computer Vision and Pattern Recognition (CVPR), 2023

Ting-Hsuan Liao, Huang-Ru Liao, **Jie-En Yao, et al., "ELDA: Using Edges to Have an Edge on Semantic Segmentation Based UDA"**, accepted by British Machine Vision Conf. (BMVC), 2022

Work Experience

Industrial Technology Research Institute

Data Science Intern at Computational Intelligence Technology Center

Jul – Sep 2020

Hsinchu, Taiwan

- Performed data preprocessing, feature engineering, and trained predictive Machine Learning models to forecast production quality, enabling data-driven quality control

Teaching Experience

Teaching Assistant of Principles of Programming for Data Science (Graduate level)

Aug - Dec 2025

- Led weekly lab sessions and office hours, supporting 50+ students develop Python and data science foundations

Teaching Assistant of Introduction to Programming (Undergrad level)

Jan - Jun 2021

- Supported curriculum organization and provided TA session to help students strengthen their C++ programming proficiency

Skills

Machine Learning / Deep Learning : PyTorch, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn

Programming Language : C, C++, Python, SQL

Software Frameworks & Tools : Docker, Kubernetes, React, ROS

Parallel Computing : MPI, Pthread, OpenMP, CUDA